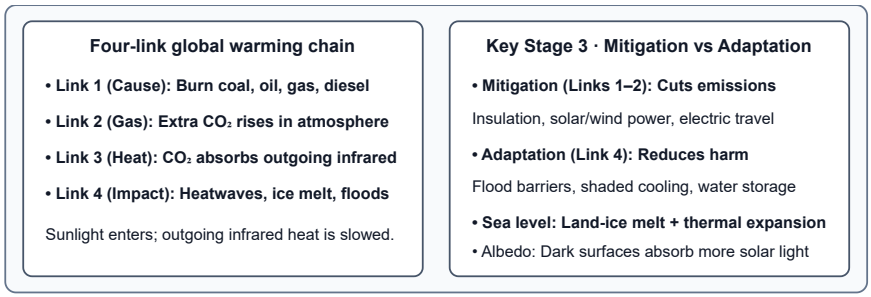
Reference sheets

Reference A · Four-link global warming chain and climate choices

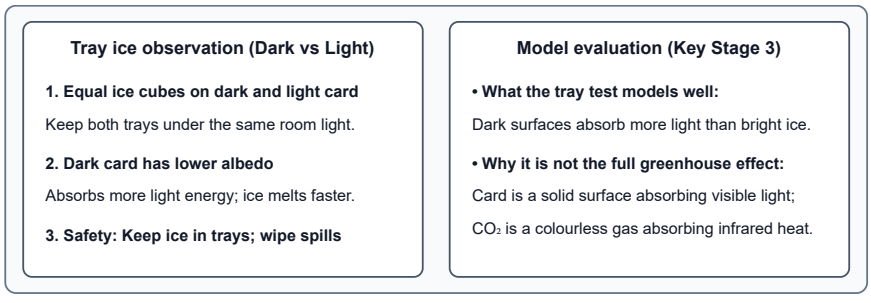


Four-link warming chain drawing. It shows fossil fuel burning and CO_2 infrared heat. It shows mitigation and adaptation.

Reference only. Do not write on this sheet. Follow Links 1 to 4 from cause to impact. Check which link each choice targets.

Chain link or choice.	What happens.	Key Stage 3 science mechanism.	Examples and limit.
Link 1 · Cause.	Power plants, heaters, and cars burn fossil fuels like coal, oil, and gas.	Burning fuel joins carbon with oxygen to make carbon dioxide (CO_2) gas.	Main human cause of extra greenhouse gas release.
Links 2 & 3 · Air & Heat.	Air CO_2 rises. More outgoing heat stays in the air.	Sun rays pass in. CO_2 and methane trap outgoing infrared heat.	The enhanced greenhouse effect slows heat leaving for space.
Link 4 · Climate impacts.	Global mean heat rises over 30 years or more.	Warmer air and seas bring heatwaves, floods, land-ice melt, and warm sea swelling.	Melting bright ice drops albedo so dark ground takes in more light.
Mitigation vs Adaptation.	Mitigation cuts gas at Links 1–2. Adaptation cuts harm at Link 4.	Mitigation: loft wraps, solar or wind power. Adaptation: flood gates, shade, water butts.	Limit: CO_2 stays in air for decades. Flood gates do not cut CO_2.

Reference B · Tray ice test (Dark card vs Light card) and model limits



Tray ice test drawing. It compares ice cubes on dark card and light card. It shows safety rules and limits.

Test feature.	**What we see on the tray.**	**Key Stage 3 reason and model limit.**
Equal ice on dark vs light card.	The ice on dark card melts faster under the same room light.	Dark card has lower albedo. It takes in more light and warms faster.
Why it is a partial model only.	It shows how dark ground takes in more sun rays when ice melts.	Limit: Dark card taking in light is not clear CO_2 gas trapping infrared heat.
Safety in the room.	Keep ice in the trays on the desk. Use room light, not a heater.	Melted ice can make a wet slip risk. Wipe any spill right away.

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use the arrival drawing. Why is burning fossil fuel at Link 1 placed before extra carbon dioxide at Link 2?

Help. Put the cause (burning fuel) before the gas it makes.

2. Look at the middle box of the arrival drawing. What happens to outgoing heat when greenhouse gases rise in the air?

Help. Follow the arrows showing heat held back in the air.

3. How is weather on one day different from long-term climate?

Help. Compare one wet Tuesday with a 30-year heat trend.

4. Look at the right box of the arrival drawing. Why do cutting fuel use and fitting flood gates act at different links?

Help. Check if each choice stops the gas or blocks the final harm.

You do tasks

1. Use Reference A. Explain two joined links in the warming chain. How does burning fossil fuels change the air? How do extra greenhouse gases warm Earth? **Staff I ☐ S ☐ C ☐**

Help. Use the words fossil fuel, carbon dioxide, greenhouse gas, and infrared heat.

2. Compare one mitigation action (loft wraps) with one adaptation action (flood gates or shade). State which link each action targets. Why do we need both? **Staff I ☐ S ☐ C ☐**

Help. Check the Link and Limit columns in Reference A.

3. Use Reference B. In the tray test, ice on dark card melts faster than ice on light card. What does this show about dark and light card? Why is it not a full model of air greenhouse gases? **Staff I ☐ S ☐ C ☐**

Help. Compare card light taking (albedo) on Reference B with air gases on Reference A.

Optional participation-loop prompt. Share your four-link chain or choice note with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. List the four links of the warming chain in order from cause to impacts.

2. Compare reducing fossil fuel use (mitigation) with a flood barrier (adaptation). Which link does each action change?

3. Why is our dark-card and light-card tray ice test only a partial model?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt